THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1 & 2: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** variable we control	Input of the experiment
2	**(d)** time taken to puff up	A number in seconds
3	**(c)** systematic investigation	Core idea of the chapter
4	**(a)** type of flour	The variable being tested
5	**(a)** investigation	Investigative world
6	**(b)** nucleus	Control centre
7	**(b)** plant cells	They contain chlorophyll
8	**(c)** Amoeba	A protozoan
9	**(a)** tissue	Level above cell
10	**(b)** bread and idli	Produces gas that makes dough rise

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Does flour type change puffing? (1). Does oil temperature change puffing time? (1).	1+1
12	It is an opinion (1). It cannot be measured or tested objectively (1).	1+1
13	Viruses reproduce only inside host cells (1). Bacteria are cells that can live and reproduce on their own (1).	1+1
14	A microalga (1). Used as a nutritious food supplement (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Yes/no: does the sugar dissolve? (1½). Measurement: time taken in seconds (1½).	1½+1½
16	Plant cell (1). Cell wall present (1). Chloroplasts present (1).	1+1+1
17	A tested prediction (1). Measurements and records from fair trials (1). Evidence to support the claim (1).	1+1+1
	OR: Root: a deep foundation of knowledge and heritage (1½). Kite: curiosity taking flight to explore new ideas (1½).	1½+1½
18	Batter contains microbes such as yeast (1). They release carbon dioxide gas (1). Gas bubbles make the batter rise and become spongy (1).	1+1+1
	OR: Bacteria are all unicellular; fungi may be uni or multicellular (1). Bacteria lack a defined nucleus; fungi have one (1). Examples: Lactobacillus and yeast (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Trials at several temperatures with same dough (2). (ii) Time to puff decreases as temperature rises (1½). (iii) Repeat, check other factors and revise the claim (1½).	2+1½+1½
	OR: Answers open new questions (1½). Puri puffing is not fully understood (1). Ideas change with new evidence (1½). Science keeps improving (1).	1½+1+1½+1
20	Rhizobium lives in nodules of legumes (1). It converts atmospheric nitrogen to usable form (1½). Soil gains nitrogen (1). Next crop grows better with less fertiliser (1½).	1+1½+1+1½
	OR: Beneficial: curd, bread, nitrogen fixation (2½). Harmful: diseases, food spoilage, crop diseases (2½).	2½+2½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1 & 2: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** phases of the Moon	Repeating lunar cycle
2	**(b)** can remain liquid	Habitable zone
3	**(b)** a testable expectation about what will happen	Tested by experiment
4	**(b)** thermometer	Measures °C
5	**(a)** Checking claims with observation	Evidence-based thinking
6	**(b)** fixing atmospheric nitrogen	Increases fertility
7	**(d)** methane	Burns to give heat
8	**(b)** microscope	Magnifies tiny objects
9	**(d)** Rhizobium causing curd formation	Rhizobium does not form curd
10	**(b)** plant cells	Stores water and substances

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Do puris puff better with fresh dough? (1). What if a hole is pricked before frying? (1).	1+1
12	The repeating phases were noticed (1). This led to the first calendars (1).	1+1
13	Microorganisms that break down dead plants and animals (1). Bacteria or fungi (1).	1+1
14	A fuel made by microbial breakdown of organic waste (1). Methane (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use identical dough circles (1). Drop one vertically and one at an angle in the same oil (1). Compare puffing and time (1).	1+1+1
16	It is a microalga rich in nutrients (1). It grows quickly (1). It is used as a food supplement (1).	1+1+1
17	To avoid burns and injury (1). Use adult help (1). Keep distance from splattering hot oil (1).	1+1+1
	OR: Grade 6: wonder (1). Grade 7: science evolves (1). Grade 8: investigation combines both (1).	1+1+1
18	Each cell type does a different job (1). Nerve cells are long to carry messages (1). Red blood cells are disc-like to carry oxygen (1).	1+1+1
	OR: Hold with both hands (1). Start with low power (1). Do not touch lenses with fingers (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Change thickness: observe puffing and thin side (1½). Change flour type: observe puffing and time (1½). Change dropping method: observe how it puffs (2).	1½+1½+2
	OR: Curiosity, careful observation, fair testing, recording, sharing results (any five) (2). Explanation of any two with example (3).	2+3
20	Green and unicellular organisms like Amoeba, Paramecium, algae (2). Put a drop of water on slide (1). Cover with coverslip without air bubbles (1). Observe first on low power (1).	2+1+1+1
	OR: They feed on dead plants and animals (1). Break them into simple substances (1). Nutrients return to soil (1). Without them wastes would pile up (1). Nutrients would not be recycled (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1 & 2: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** something we measure	Observed or measured outcome
2	**(b)** the same thickness	Keep other factors the same
3	**(c)** a place for scientific observation	Science is everywhere
4	**(c)** way the dough enters the oil	A controllable factor
5	**(a)** a deep foundation of knowledge and heritage	Rooted in environment and traditions
6	**(a)** cytoplasm	Fills the cell
7	**(b)** chloroplast	Contains chlorophyll
8	**(b)** fungi	Yeast is unicellular, mushroom is multicellular
9	**(b)** organ	Made of several tissues
10	**(c)** carbon dioxide	Inflates the balloon in the activity

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It checks whether results are consistent (1). It reduces chance errors (1).	1+1
12	Observation is what we see or measure (1). Prediction is what we expect to happen (1).	1+1
13	Shape suits the function (1). Example: long nerve cells carry messages (1).	1+1
14	Gives shape and support (1). Protects the cell in plants, fungi and bacteria (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It gives numbers to compare results (1). It makes observations precise (1). It allows others to repeat the test (1).	1+1+1
16	Breakdown of sugar by microbes without oxygen producing gas or acid (1½). Used in making bread, idli or curd (1½).	1½+1½
17	It shows the same method of observation helps (1). Humans influence climate (1). Science can guide actions to protect Earth (1).	1+1+1
	OR: How does the lit part change from night to night? (1½). How many days does a full cycle of phases take? (1½).	1½+1½
18	An English scientist (1). He observed a thin slice of cork under a microscope (1). He saw small boxes he called cells (1).	1+1+1
	OR: All living bodies are made of cells (1). Cells perform life activities like taking in food and energy (1). Cells come from existing cells (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Opinion cannot be tested, e.g. puris are tasty (1½). Scientific statement can be tested, e.g. hotter oil puffs puris faster (1½). Evidence settles scientific statements (2).	1½+1½+2
	OR: Variable: water temperature (1). Observation: time to dissolve (1). Fixed: amount of sugar, volume of water, stirring (1). Use same quantities each time (1). Record in a table (1).	1+1+1+1+1
20	Cell wall outside and membrane inside (1). Cytoplasm inside (1). Nucleoid with genetic material (1). No membrane-bound nucleus (1). Hence a simple cell (1).	1+1+1+1+1
	OR: He developed oil-eating bacteria (1½). They break crude oil (1). Help clean oil spills (1). Decomposers clean waste (1). Reduce pollution (½).	1½+1+1+1+½

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** a smaller lit part is visible from Earth	Phases depend on relative positions
2	**(a)** climate patterns	Climate change
3	**(b)** reliable	Repetition checks consistency
4	**(b)** not completely understood by scientists	Science keeps exploring
5	**(c)** calendar based on sky observations	Calendars link to Sun and Moon
6	**(b)** decomposers	Clean up the environment
7	**(b)** Ananda Mohan Chakrabarty	Cleaning oil spills
8	**(a)** Robert Hooke	He used an early microscope
9	**(b)** almost everywhere, even inside other organisms	Found in all environments
10	**(d)** Lactobacillus – curd	Roles of microbes

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Without a unit the number has no meaning (1). Example: 5 s differs from 5 min (1).	1+1
12	Many everyday events can be investigated (1). No special laboratory is needed (1).	1+1
13	Cause diseases in humans, plants and animals (1). Spoil food (1).	1+1
14	Bacteria that break down crude oil (1). They help clean up oil spills (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fair: only oil temperature is changed (1½). Unfair: temperature and thickness both changed so cause is unclear (1½).	1½+1½
16	Diseases like cholera (1). Food spoilage (1). Plant diseases reducing crops (1).	1+1+1
17	Living things depend on air, water, sunlight and each other (1½). Understanding them helps protect life (1½).	1½+1½
	OR: Does a puri puff with different flours? (1). How quickly does sugar dissolve in warm and cold water? (1). How does the Moon's shape change over days? (1).	1+1+1
18	Nucleus controls work like a manager (1). Cytoplasm is the workplace (1). Membrane controls what enters and leaves like a gate (1).	1+1+1
	OR: Yeast – bread making (1). Lactobacillus – curd formation (1). Rhizobium – nitrogen fixation (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Questions can be asked about everyday events (1). Factors can be changed and controlled (1). Observations can be recorded (1). Method is the same as in big experiments (1). Science needs curiosity not a fancy lab (1).	1+1+1+1+1
	OR: Observations and measurements show temperature changes (2). Experiments and models explain causes (1½). Findings guide actions to protect Earth (1½).	2+1½+1½
20	Viruses are much smaller (1). They reproduce only inside hosts (1). Do not grow outside (1). Not made like a cell (1). Cause diseases (1).	1+1+1+1+1
	OR: Bread (1). Curd (1). Soil fertility (1). Biogas (1). Cleaning waste or oil spills (1).	1+1+1+1+1

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